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AI engineer who ships LLM systems end to end — hybrid retrieval with reranking and measured evaluation, multimodal model training, and FastAPI services running on constrained infrastructure. B.Tech Computer Science, 2026 graduate. Strongest work is a production-grade RAG system with measured retrieval quality, and ETL plus time-series forecasting pipelines built during an IBM internship.

TECHNICAL SKILLS

LANGUAGES	Python · SQL · TypeScript · C++
AI / ML	LLMs · Retrieval-Augmented Generation · Fine-tuning (LoRA, QLoRA) · NLP · Computer Vision · Multimodal · Time-Series Forecasting · Embeddings
FRAMEWORKS	PyTorch · TensorFlow · Hugging Face Transformers · PEFT · scikit-learn · OpenCV · React / Next.js · Pandas
INFRA	FastAPI · Docker · Render · MLflow · Git · Gemini, OpenAI & Groq APIs
DATA STORES	Pinecone · ChromaDB · Supabase · MongoDB

EXPERIENCE

Research Intern — Visys · research collaboration with a Manchester (UK) researcher Jul 2026 – Present

- Building ARMS, a localise-first ship inspection pipeline — DINOv2 retrieval across a 150-view render bank, keypoint correspondence and solvePnP camera-pose refinement, and an abstain gate that rejects untrustworthy poses instead of guessing forward.
- Report each of 42 catalogued CAD parts as present, shifted, rotated or missing by projecting them onto the solved pose — making absence measurable, which a bounding-box detector cannot do. Python, typed with mypy, 41 tests passing.

AI / ML Intern — IBM · ExpensexchangeAI project Jun – Jul 2025

- Built ETL pipelines and time-series forecasting models for a stock portfolio analyser covering 20+ tickers — 76% directional accuracy, R^2 0.82.
- Cut forecast error 7–10% through Optuna hyperparameter tuning, Huber loss optimisation and domain-specific feature engineering.
- Ran MLOps workflows in MLflow for experiment tracking, model versioning and staging-to-production deployment.

PROJECTS

RAG Lab — Production RAG Document Q&A System [/RAG-LAB](#) · [live demo](#) Jan 2026 – Present

- Built hybrid dense + sparse retrieval over multi-format documents — Pinecone semantic embeddings alongside Supabase full-text and BM25 lexical search, unified by Reciprocal Rank Fusion and reranked with a local CrossEncoder.
- Built a standalone LLM-as-a-judge evaluation suite scoring Groundedness, Context Relevance and Answer Relevance from 0–1 with written reasoning; the end-to-end integration suite reports 0.95 / 0.90 / 0.95 on its sample corpus.
- Cut server memory from over 1 GB to under 100 MB by moving embeddings from a local BAAI/bge model to the Gemini embedding API, letting the system run inside a 512 MB free tier. Next.js + React frontend, FastAPI REST API, Dockerised and deployed on Render.

MELD Emotion — Multimodal Emotion Recognition [/Human-Emotion-Detection-System](#) Jan – May 2026

- Trained a three-modality pipeline — Llama-3.2-3B with LoRA/PEFT (text), CNN-BiLSTM on Mel-spectrograms (audio), 3D ResNet (video) — trained on 9,816 samples across 7 emotion classes on an HPC GPU cluster, fused by TA-AVN cross-modal attention with Squeeze-Excitation gating.

Echo-D — Obstetric Ultrasound Diagnostics Aug – Dec 2025

- Built an offline diagnostic pipeline for placenta accreta spectrum — SAM zero-shot segmentation, BLIP-2 radiologist-style captioning and Grad-CAM++ interpretability, scored on Dice, IoU, AUC and F1; plus a synthetic ultrasound generator to offset medical data scarcity.

NextAIQLabs — AI Skills Platform for Indian Schools Jan 2024 – Present

- Built a schools platform (app + portal) connecting schools, teachers, parents and administrators, with end-to-end programme tracking and collaboration.
- Designed a practical AI curriculum for Classes 6–12 modelled on Elon Musk's Ad Astra school; ran discovery with decision-makers at 4 schools and produced a costed operating model.

EDUCATION & ACHIEVEMENTS

B.Tech, Computer Science & Engineering (AI & ML) — UPES, Dehradun · CGPA 7.26 / 10 Aug 2022 – Jul 2026

Coursework: Machine Learning · LLMs & Generative AI · Computer Vision · Explainable AI · Data Structures & Algorithms

- 2nd Runner-Up, IBM Hackathon — end-to-end AI solution design under competitive evaluation.
- UPES DSW Ambassador, Jan 2024 – Feb 2025 — student-administration liaison and campus programme delivery.